ASSIGNMENT-11

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BATCH:29

CODE:

```
class FabricNetwork:
    def __init__(self):
        self.ledger = {}

    # Create Asset
    def create_asset(self, asset_id, owner):
        self.ledger[asset_id] = {"ID": asset_id, "Owner": owner}
        print("Asset Created (Fabric) ✅:", self.ledger[asset_id])

    # View Ledger
    def view_assets(self):
        print("\nFabric Ledger State 📜:")
        for asset in self.ledger.values():
            print(asset)

    # Transfer Ownership
    def transfer_asset(self, asset_id, new_owner):
        if asset_id in self.ledger:
            self.ledger[asset_id]["Owner"] = new_owner
            print("Ownership Transferred (Fabric) ✅:", self.ledger[asset_id])
        else:
```

```
        print("Asset not found ❌")
# Simulating Ethereum (just for comparison)
class EthereumNetwork:
    def transfer(self, sender, receiver, amount):
        gas_fee = 0.01
        print("\nEthereum Transaction ⛓️")
        print("From:", sender, "To:", receiver)
        print("Amount:", amount, "ETH")
        print("Gas Fee:", gas_fee, "ETH ❌ (High)")
        print("Status: Slower Confirmation ⏳")
# Execution
fabric = FabricNetwork()
eth = EthereumNetwork()
# Fabric operations
fabric.create_asset("A1", "AKSHU")
fabric.create_asset("A2", "VINNU")
fabric.view_assets()
fabric.transfer_asset("A1", "VINNU")
fabric.view_assets()
# Ethereum comparison
eth.transfer("AKSHU", "VINNU", 5)
```

OUTPUT:

```python
class FabricNetwork:
    def __init__(self):
        self.ledger = {}

    # Create Asset
    def create_asset(self, asset_id, owner):
        self.ledger[asset_id] = {"ID": asset_id, "Owner": owner}
        print("Asset Created (Fabric) ✅:", self.ledger[asset_id])

    # View Ledger
    def view_assets(self):
        print("\nFabric Ledger State 📜:")
        for asset in self.ledger.values():
            print(asset)

    # Transfer Ownership
    def transfer_asset(self, asset_id, new_owner):
```

```
Asset Created (Fabric) ✅: {'ID': 'A1', 'Owner': 'AKSHU'}
Asset Created (Fabric) ✅: {'ID': 'A2', 'Owner': 'VINNU'}

Fabric Ledger State 📜:
{'ID': 'A1', 'Owner': 'AKSHU'}
{'ID': 'A2', 'Owner': 'VINNU'}
Ownership Transferred (Fabric) ✅: {'ID': 'A1', 'Owner': 'VINNU'}

Fabric Ledger State 📜:
{'ID': 'A1', 'Owner': 'VINNU'}
{'ID': 'A2', 'Owner': 'VINNU'}

Ethereum Transaction ⛓️
From: AKSHU To: VINNU
Amount: 5 ETH
Gas Fee: 0.01 ETH ❌ (High)
Status: Slower Confirmation ⏳
PS C:\Users\nampa\OneDrive\Blockchain>
Ethereum Transaction ⛓️
From: AKSHU To: VINNU
```